

**CC7183 DATA ANALYSIS AND
VISUALISATION**

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A Time Series Analysis of Household Expenditure Patterns (2019–2024)

Abstract— An insight into household spending assists them in comprehending the manner in which individuals utilize goods and services as well as assists in planning the economy, particularly when there are massive shifts. This paper examines the expenditure incurred by households between 2019 and 2024 using charts, which demonstrate the data over a period of time, and emphasize the shifts that occurred due to the COVID -19 pandemic. They utilized government expenditure information that is open. The data analysis was done in the usual stages of cleaning data, building models and interactive charts. They also performed a time-based analysis of the data to identify long-term tendencies. They also compared one year with another and checked certain branches in spending to know whether it was steady or variable. They discovered that the fundamental expenditure such as housing, food and transport remained relatively constant. However, expenditure on non-essentials altered more in the case of the pandemic. The results were also easier to digest using Power BI dashboards as users were able to explore trends and compare categories as they were browsing through. According to the examination, time-series examination combined with graphical representations is an effective method to clarify the complicated expenditure patterns. This contribution to the literature provides multiple-year analysis which is presented in images, and it provides practical thoughts to policy makers, families, and prospective research on consumption by people during periods of economic uncertainty.

Keywords— Household Expenditure, Time Series Analysis, Data Visualisation.

I. INTRODUCTION

The analysis of household spending is highly necessary in the realization of how individuals conduct themselves economically as well as in coming up with effective policies and budgeting. The expenditure of people gives helpful data on the quality of life, the difficulty of affording the daily requirements, and the allocation of money on necessities and luxury goods. Spending data are used by governments, policy makers, and economists alike to determine the stability of the economy, to coordinate social programs as well as anticipate future needs. Thus, the analysis of household expenditure over time is an essential component of the research of the big picture economy and society functioning [1].

The COVID-19 epidemic was a massive disturbance to the globe and the national economies. It disarranged employment, decreased stability of incomes, and reduced the confidence of people in purchasing. Regulations that prevented individuals to move around, novel forms of employment and future concern altered the manner in which households managed monetary assets. Those changes explain why they should consider spending prior to, throughout, and following the pandemic to understand how families adapt to major economic shocks. A good method to monitor the change of spending patterns with time is by time-series analysis. It aids them in discovering tendencies, good and bad, and the way they come back to ourselves in a few years. Using it with such tools as Power BI, they are able to demonstrate this hard economic data in a clear and understandable way and allow people to interact with it. The proposed research is expected

to examine the household spending in the period of 2019–2024 through time-series visual analytics. They would like to know how the COVID-19 influenced purchasing patterns and note the essential trends that can aid individuals in making better economic choices [2].

II. LITERATURE REVIEW

A. Household Expenditure Analysis in Economic Research

Economic researches often use household spending studies in understanding the state in which people live, what they purchase, and the stability of their finances. The way people use their money on essentials and luxury demonstrates where they invest their earnings and therefore it is a significant indicator of making regulations and designing the economy. According to Madudova and Corejova, one of the most important aspects of the big-picture economics is the ability to measure household expenditures precisely, as it demonstrates how economies transform and what taxes and other money regulations to select. The approaches of the old studies involved massive amounts of spending data to examine long-term profiles but the approaches encounter issues in comparing the data, biases associated with the combination of numbers and real-time tracking of spending. The above issues demonstrate that there is a necessity of powerful analysis instruments, which would be capable of following changes in household expenditure at any given period of time [3].

B. Impact of COVID-19 on Consumer Spending Behaviour

The COVID-19 outbreak brought substantial transformations in the way individuals purchase goods globally, thereby leaving researchers extremely inquisitive about the impacts of this pandemic on the economy. J.P. Morgan attributes the shift in consumer spending to restrictions and uncertainty throughout the pandemic. Consumers purchased more necessities and minimized consumption on travelling, recreational activities and restaurants. Sharma demonstrates that people became more cautious, thought that money and illness worry make them buy, emphasize on the essentials, and save more during the pandemic. Comparing families in the United States and Israel, Roll and colleagues discovered that the COVID-19 increased financial straining and compelled families to redirect spending when jobs and income were at risk. Combined, these studies are capable of being used to affirm that COVID -19 was a significant shock that fundamentally altered the household consumption behavior [4].

C. Time-Series Approaches in Consumption Analysis

The tendency to observe how the household spending varies with time is a popular method of time-series analysis. It assists in determining the patterns, ups and downs and recovery periods. According to Madudova and Corejova, such a time-based analysis is particularly useful in revealing large shifts in response to shocking external events, such as economic crises or pandemics. Research that examines the

impact of COVID-19 on expenditure tends to compare pre-pandemic and post-pandemic spending of people. This method provides helpful hints as to how the spending patterns vary during big shocks of the economy.

D. Role of Data Visualisation and Research Gaps

Even though there is increased literature about household spending and COVID-19, there remain big gaps. Most of the research incorporates only plain table or simple statistics and it is difficult to check how the spending evolves over time. Charts are beginning to be utilized in industry reports, however the academic work hardly utilizes interactive charts that can aid people in comprehending the information and making choices. In addition, majority of the studies examine only short-term impacts of the pandemic rather than long-term and multi-year patterns. This work seals such gaps because time-series analysis and interactive displays are employed to understand household spending during the period 2019-24 in greater detail and with valuable information [6].

III. METHODOLOGY

To conduct this study, they have used Data Analytical Life Cycle, which is an easy-to-follow six-step model that assists them to locate problems, manipulate data, develop models, and distribute findings. The process has paralleled the Power BI process that they had followed and rendered the analysis transparent and reproducible.

A. Discover

The paper examines the dynamics of household spending and the impact of the COVID-19 pandemic on it. The analysis will identify the long-term trends, changes in categories of spending, and pre- and post-COVID changes. Household expenditure informs them of economic wellbeing and cost of living and therefore, this analysis is significant to the policy makers, planners and family. The aim of the analytics is to analyze the spending patterns through the years, identify significant changes, and present the findings in concise, practical visuals [7].

B. Data Preparation

The information was provided by a government source. It demonstrates the expenditure of households per year between 2019 and 2024. The key facts are the nature of expenditure, the year, and the average expenditure that was made per week. Initially they viewed the information and noticed that it was in extensive structure, whereby one column represented a year. Such a layout is not suited to time-series analysis. Therefore, they prepared the data in Power Query using Power BI. The transformation that was most important was the unpaved into one column of the year columns whereby they ended up with a long format. They also converted the column with the Year to numbers, harmonized the column names and verified the missing or incorrect values. These procedures prepared the data to be subject to time-series modelling [8].

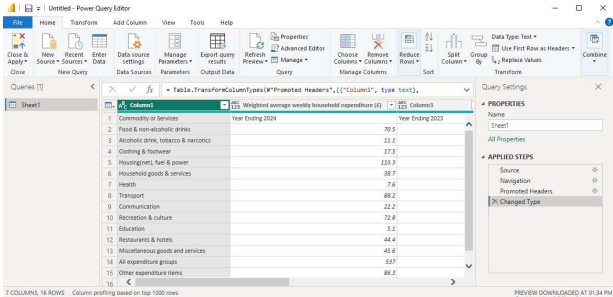


Figure 1: Before Data Cleaning & Transformation.

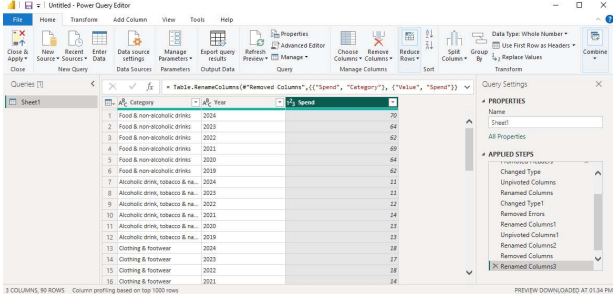


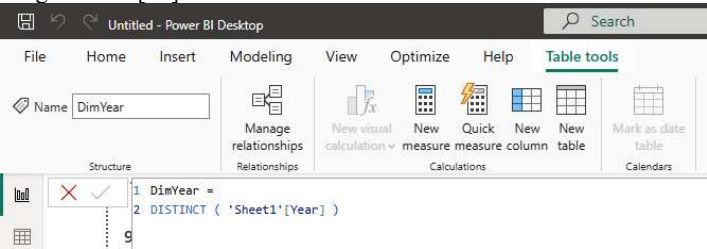
Figure 2: After Data Cleaning & Transformation.

C. Plan Model

The data spans numerous years and this is why they selected a time-series approach. It allows them to observe the way of spending with the changes in time, trends, highs and lows and the way it re-bound. Other tools, such as classification or clustering, are not as applicable in this case since they are analyzing changes over time, but not grouping or projecting categories. Time-series analysis will allow them to directly compare prior and post-disruption spending following such a big disruption as COVID-19 [9].

D. Build Model

The analysis is based on simple time series tools, such as trend analysis and year to year comparison. They created custom DAX measures in Power BI to locate the total spending and observe how it will vary annually. They also developed a simple category in which they have labeled data as pre-COVID or post-COVID to further simplify comparisons. Since the data is given in annual aggregate they did not apply machine-learning models. Rather they have preferred mere descriptive analysis so as to make it simple and easy to interpret. It is the correct decision in the field of policy work where the transparency plays a great role [10].



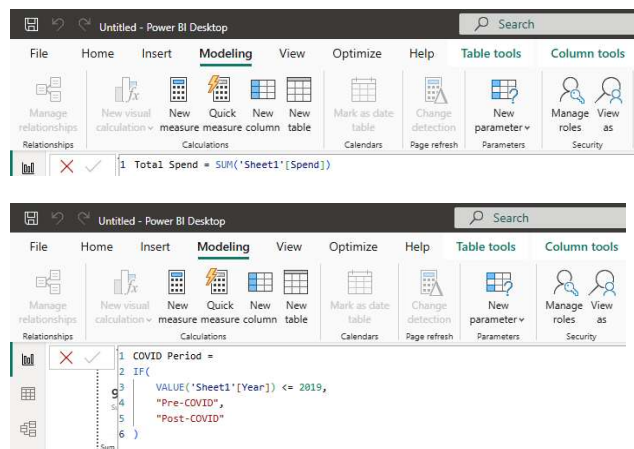


Figure 3: DAX Measures screen.

E. Communicate

Insights were shared on an interactive Power BI dashboard that assists individuals in exploring the data. Visual aids that were planned were line charts to illustrate the trends over time, bar charts to compare various categories, KPI cards to display the most important summary numbers, and slicers to filter by Year, Category, and COVID Period. Power BI has been selected as it allows combining data analysis with visualisations interactively so that users can explore how spending works in an interactive and non-commonly challenging manner [11].

F. Measure effectiveness/Apply Live

Public, non-personal data was used, and therefore there were no issues of privacy or consent. They presented the findings cautiously without making any cause claims where the evidence did not substantiate the claims. In brief, the data analytics procedure provided a good and appropriate method to address the research problem and present credible and policy-relevant information.

IV. RESULTS AND DISCUSSION

A. Model Overview and Integration with Research Objectives

Simple time-series model They were studying household expenditure spending on money between 2019 and 2024. It was not aimed at making future predictions according to categories or groups but at looking at how spending is going to change over time. They cleaned the data, aspecialized total spending and measured the year to year change in the spending. The method allows observing long-term trends, which categories of spending are most prevalent and how the COVID -19 pandemic impacted household expenses [12].

B. Time-Series Trends and COVID-19 Impact

The way the household spending varied with time is depicted in figure 4. It demonstrates that the spending was relatively stable before 2020, whereas it fluctuated up and down during the COVID-19 crisis. The onset of the pandemic resulted in different ways, in which people used their money due to uncertainty, traveling restrictions, and unstable incomes. The level of spending did not simply fall on a

straight line, but rather, people transferred the amount they spent, and they demonstrated their adaptability. Following the pandemic, the pace of spending gradually began to stabilize, partially recovering and people adapting to new economic circumstances [13].

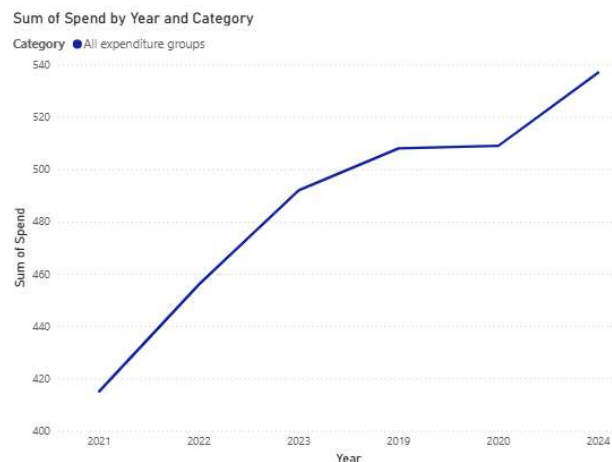


Figure 4: Time-series trends in household expenditure across categories from 2019 to 2024.

C. Category-wise Expenditure Patterns and Variability

The comparison at Figure 5 demonstrates that the largest segments of household expenditures are on the essentials of housing, food, and transport. Such expenses do not vary much and this has proven to be very important regardless of what transpire on the economy. Conversely, non- essential expenditure e.g. recreation, culture and hospitality expenditure vary year in year out. It implies that when the economy is volatile families invest in necessities and reduce or postpone non-necessity purchases. Some categories of spending have short-term increases in small odd bumps. These may be because of slow purchasing or hasty policies [14].

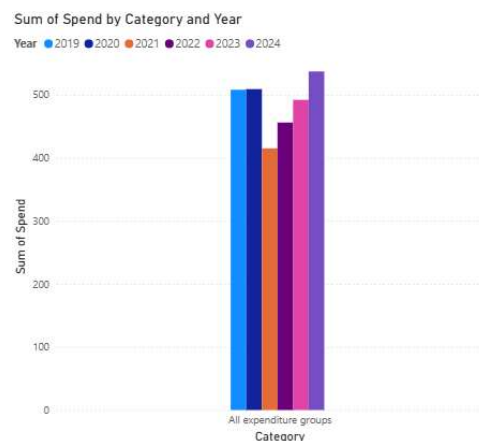


Figure 5: Category-wise household expenditure comparison highlighting stability and volatility across years.

D. Visual Interpretation and Complementary Insights

It is easier to visualize the change in spending in various angles using both line and bar charts. The line chart demonstrates the movement of the spending over time and the bar chart demonstrates the way each category is compared to others every year. They provide a complete picture together

showing both long-term trends, and category differences. No extreme outliers are present hence data appears consistent and credible [15].

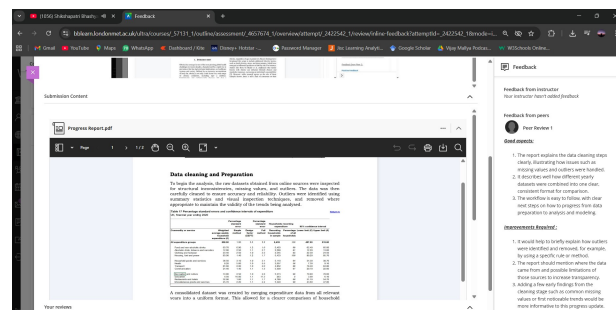
E. Practical Implications

Policymaking-wise, the findings indicate that the expenditure on essentials remains constant amid economic shocks whereas the expenditure on non-essentials is more inclined to decline. To families, the results emphasize the fact that flexible budgets need to be used. They ought to spend on necessities but still have some room of spending on non-necessities whenever there is uncertainty in times.

V. PEER REVIEW

The peer review provided positive feedback that was used to enhance transparency and general quality. The reviewer has commended the explanation of data cleaning and preparation being well explained, particularly the step-by-step flow regarding data preparation to analysis and modeling. The feedback on that proved that the workflow was consistent, well-organized, and comprehensible. Nevertheless, the reviewer also mentioned aspects that could be better: it is not clear how the outliers have been identified and managed, they should be told more about the sources of data, and it is better to mention early observations that were made during the cleaning process [18].

I put my feedback into careful consideration and used it to revise the study. This simplified the methods and provided more reasons behind my decisions in analysis. Giving feedback to another student also helped me to learn to judge the analysis steps sharper, identify my strong points and weak points, and be able to explain my recommendations clearly and usefully. In general, the peer review contributed to my further professional development as it helped me to think more, enhance my writing skills in the field of technical writing, and understand that it is crucial to be clear, open, and responsible when presenting data analytics [19].



VI. CONCLUSION AND RECOMMENDATION

This research paper has examined the expenditure of households between 2019 and 2024. The time-series visual analytics technique was applied. They were concerned with changes in the COVID-19 pandemic. The findings indicate that they have achieved the objectives. The determined some significant trends with time. They observed the most significant spending categories. They also observed the way before and after the pandemic spending was different. Basic goods such as housing and food and transport remained similar. The non-essential expenditure was more fluctuating in harsh economic conditions [16].

Interactive visualisation and time-series analysis were useful to demonstrate and describe the change in spending

over time. The Power BI dashboards provide the user to view clear trends and data across various years and categories and they will be simpler to analyze and utilize. The key advantage of the study is that it uses the visual analytics to surpass the conventional spending analysis to provide a clear, decision-making picture of how households spend in many years.

Based on this, the legislators can utilize the findings to implement specific assistance that will ensure that basic family expenditure remains secure in subsequent economic crises. Families can save themselves by implementing flexible budgets, which emphasize the mandatory expenditures and reduce unnecessary expenditure in times of uncertainty. Further future research might be developed on the basis of this research by applying more periodical data, such as monthly records of expenses, and forecasting models to estimate patterns of spending. To make the analysis more profound and convenient to work with, the models might be enhanced with scenario testing and automatic identification of unusual spending [17].

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VI. APPENDIX

(POWERBI DASHBOARD)

